Lab 3
CS30812 - Artificial Intelligence
Sultan Abdu

Task 1:

Part (1):

Row	**rain**	**hagrid**	**dumbledore**	**Not(rain) -> hagrid**	**hagrid OR dumbledore**	**Not(hagrid AND dumbledore)**	**dumbledore**	**KB True**
1	F	F	F	F	F	T	F	No
2	F	F	T	F	T	T	T	No
3	F	T	F	T	T	T	F	No
4	F	T	T	T	T	F	T	No
5	T	F	F	T	F	T	F	No
6	**T**	**F**	**T**	**T**	**T**	**T**	**T**	Yes (Model)
7	T	T	F	T	T	T	F	No
8	T	T	T	T	T	F	T	No

Part (2):

MsScarlet:

Is true because the known cards eliminate Professor Plum (Not(plum)) and Colonel Mustard (Not(mustard)), as such Ms. Scarlet is left.

Library:

Is true because the known negative cards (Not(ballroom), Not(kitchen)) rule out the ballroom and kitchen.

Knife:

Is true because the revolver and wrench exclude the initial constraints (Not(revolver)).

Task 2:

Part (1):

Prediction (a):

If you remove dumbledore from harry.py, the program will now print False.

Result (c):

Without dumbledore rain is no longer true on every valid model so it returns False.

Part (2):

Prediction (a):

If you delete Symbol("MinervaGryffindor") from puzzle.py it removes the fixed Minerva's house. Otherwise multiple people can cross houses, making house placements ambiguous.

Result (c):

Outputs fewer or no symbols because multiple valid models are satisfied leaving the houses ambiguous.

Adding a new unique clue (such as knowledge.add(Symbol("MinervaRavenclaw"))), It becomes constrained. Printing a unique house again for every person.